

## Review

# Comparative efficacy of combined exercise and nutritional interventions for sarcopenia: A systematic review and network meta-analysis incorporating remote delivery models

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## HIGHLIGHTS

- Applying rigorous network meta-analysis methodology, this review systematically compares interventions for sarcopenia, though the GRADE analysis reveals limited certainty in the available evidence.
- Resistance exercise, when combined with other forms of physical activity and nutritional support, proves to be more effective for addressing sarcopenia than resistance exercise alone.
- Our analysis clarifies that nutritional supplementation alone is insufficient, redirecting focus towards synergistic multimodal approaches and highlighting the need for future research on personalized nutrition protocols within combination therapies.

## ARTICLE INFO

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Sarcopenia

Network meta-analysis

Comprehensive comparisons

Non-pharmacological interventions

## ABSTRACT

**Objective:** To identify the most effective interventions to delay the progression of sarcopenia and potentially improve muscle strength and mass, with the ultimate goal of reversing sarcopenia, by comparing all pharmacological and non-pharmacological interventions.

**Design:** Systematic review and network meta-analysis.

**Findings:** This analysis identified 115 randomized controlled trials (RCTs) with 10,967 participants which an average age of 74.85 years, together evaluating 27 different interventions. For the primary outcome, endurance training followed by resistance training yielded the highest effect on muscle strength (SMD 2.59, 95% CI 1.50 to 3.69), while general remote training was most effective for lean mass (SMD 1.35, 95% CI 0.41 to 2.29) and physical performance (SMD 2.39, 95% CI 1.26 to 3.53). The combination of nutrition with aerobic and resistance training significantly improved muscle strength (SMD 1.25, 95% CI 0.64 to 1.87) and physical performance (SMD 1.73, 95% CI 1.09 to 2.37). Resistance training significantly improved multiple secondary outcomes: body fat mass (SMD -0.57, 95% CI -0.75 to -0.40), blood inflammatory markers (SMD -1.69, 95% CI -1.95 to -1.43), and knee extension strength (SMD 0.55, 95% CI 0.33 to 0.76). Aerobic exercise reduced body fat (SMD -0.48, 95% CI -0.87 to -0.09) and inflammation (SMD -1.98, 95% CI -2.31 to -1.66), while improving quality of life (SMD 0.71, 95% CI 0.53 to 1.03). Subgroup and meta-regression analyses identified sample size, age, and intervention duration as significant moderators of treatment effects, while sensitivity analyses confirmed the robustness of findings to statistical methods.

**Conclusions:** This network meta-analysis concludes that applying a multimodal approach involves combining resistance training, aerobic exercise, and targeted nutrition to the sarcopenia population, which will be the most effective strategy for managing sarcopenia. Future efforts should focus on enhancing remote intervention delivery to improve accessibility and adherence among older adults.

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## 1. Introduction

Sarcopenia is consistently defined as a progressive and generalized skeletal muscle disorder (Cruz-Jentoft et al., 2019) and has been assigned an ICD-10-CM Diagnosis Code. Although there is currently no universally accepted global definition of sarcopenia, two authoritative consensus definitions exist: the European Working Group on Sarcopenia in Older People 2 (Sayer et al., 2024) and the Asian Working Group for Sarcopenia 2019 (L. K. Chen et al., 2020), including the above two diagnostic criteria. Alongside these, others widely cited diagnostic criteria for sarcopenia generally encompass three key aspects: low muscle strength, low muscle mass, and low physical performance. Although there are differences in the specific diagnostic modalities and thresholds for each parameter, variations in specific diagnostic modalities and thresholds may be attributable to regional variations (Z. Chen et al., 2025). The prevalence of sarcopenia among the general population worldwide is estimated to be 10% to 16% (Yuan & Larsson, 2023). The progression of sarcopenia can be accelerated by comorbidities such as lung cancer (Lyu et al., 2025), especially when these conditions are difficult to manage or persist over the long term. Given the pathophysiological change characteristic of muscle aging, nutritional intake is essential for managing both age-related muscle loss and sarcopenia (Calvani et al., 2023). With the global aging population, the prevalence of sarcopenia is expected to increase, leading to adverse outcomes such as falls, functional decline, frailty, and impaired recovery from other illnesses, and increased mortality. As a geriatric syndrome marked by declines in skeletal muscle strength and function, sarcopenia has attracted considerable research interest, prompting investigations into strategies that may slow its progression (Yang, Liu et al., 2025).

Although sarcopenia predominantly affects the elderly, its manifestations are not exclusively confined to this population (Mager, Hager & Gilmour, 2023). In the existing intervention studies (Shen et al., 2023; Yang, Pan, Luo, Liu & Ossowski, 2025), non-pharmacological interventions have been explored, including resistance exercise, aerobic exercise, multi-modal exercise regimens, and nutritional supplementation (e.g., vitamin D, protein, and amino acids). Additionally, some researchers have undertaken intervention studies involving pharmacological agents, such as Traditional Chinese Medicine (TCM) (Xie et al., 2025), hormones (Gentile et al., 2025), and calcium supplements (Moreira-Velasco, Contreras-Alvarado, Rammal, Rivas & Duque, 2025). While both narrative and systematic reviews have established the efficacy of the Mediterranean diet in improving sarcopenia, comparative evidence on the relative effectiveness of single nutrients, whole foods, and overall dietary patterns remains absent (Granic, Sayer, Cooper & Robinson, 2025). A systematic review (Shen et al., 2023) shows that resistance exercise, whether combined with nutritional support or aerobic and balance training, yields the greatest improvement in quality of life. However, no definitive conclusions can be drawn regarding its superiority for increasing muscle strength. The optimal outcome measures for evaluating intervention effectiveness remain the three core diagnostic criteria for sarcopenia. Notably, none of these interventions have gained consensus among experts or received formal drug approval.

Current evidence on sarcopenia interventions remains fragmented. Existing systematic reviews are often limited by scope, focusing solely on exercise, nutritional, or other single-domain interventions, and by language, frequently including only English-language publications. These limitations hinder cross-comparison of findings and reduce their practical utility for clinical decision-making. To address this gap, we will conduct a comprehensive network meta-analysis. This study aims to synthesize direct and indirect evidence across multiple intervention types, thereby generating hierarchically ranked, clinically actionable findings. Our results are intended to inform practice guidelines and support the development of effective, integrated strategies for sarcopenia management and prevention.

## 2. Methods

The protocol for this systematic review and network meta-analysis was prospectively registered with PROSPERO (registration number: CRD420251066517). This study was conducted and reported in accordance with the PRISMA (preferred reporting items for systematic reviews and meta-analyses) 2020 (Page et al., 2021) and the PRISMA extension for Network Meta-Analyses (Hutton et al., 2015).

### 2.1. Literature search strategy

A comprehensive literature search was performed across the following databases up to March 13, 2025: CINAHL, The Cochrane Library, EBSCO, Embase, Medline, PEDro, PubMed, SinoMed, Web of Science, CNKI, WanFang Database, and VIP Information. The detailed search strategy is provided in the Supplementary Methods 1.

### 2.2. Inclusion and exclusion criteria

The studies included in this research adhere to the following criteria: (1) Population: Participants diagnosed with sarcopenia, irrespective of the specific diagnostic criteria employed; (2) Intervention: All forms of intervention related to sarcopenia were considered; (3) Control group: Placebo, standard care, or any active intervention used as a control; (4) Outcomes: Primary outcomes were defined according to the most widely used diagnostic criteria for sarcopenia, including muscle strength, lean body mass, and physical performance. Secondary outcomes were selected based on frequency of reporting, including body fat mass, quality of life, blood biomarker, and knee extension strength; (5) Study design: The intervention must be conducted as a randomized controlled trial.

Exclusion criteria were as follows: (1) Duplicate publications without original data, such as reviews, protocols, animal studies, and conference abstracts; (2) Articles not published in English or Chinese; (3) Chinese articles not indexed in the Chinese Science Citation Database (CSCD). CSCD-indexed publications are regarded as a benchmark for high-quality research in Chinese natural sciences, adhering to stringent editorial and review processes. To maintain the high quality and reliability of our study, only articles indexed in the CSCD were included.

### 2.3. Study selection and data extraction

All retrieved records were imported into EndNote version 9. Two independent reviewers executed a two-phase study selection process. Initially, all records were screened by title and abstract, and those considered potentially eligible by either reviewer proceeded to full-text retrieval. In the second phase, the same reviewers independently assessed the full texts against prespecified eligibility criteria, with conflicts resolved through consensus or consultation with a third reviewer if necessary. Throughout this process, the two reviewers worked independently without communication. After completing the full-text review, their results were cross-checked. Two reviewers independently extracted data using a pre-piloted form. Information collected included: first author, publication year, country of origin, sample size, age, sex of participants, intervention details, population control sources, trial duration, diagnostic criteria employed, outcomes measured, and tools used for outcome measurement etc. A summary of the literature screening and selection process is provided in Fig. 1.

### 2.4. Quality appraisal

The risk of bias for each included study was independently assessed by two reviewers using the original Cochrane Risk of Bias Tool (RoB 1.0) for randomized trials (Higgins et al., 2011). This established tool evaluates bias across seven key domains: random sequence generation, allocation concealment, blinding of participants and personnel, blinding

of outcome assessment, incomplete outcome data, selective reporting, and other sources of bias. For each domain, individual judgments were categorized into one of three predefined levels: “low risk,” “high risk,” or “unclear risk.” Review Manager (RevMan) v. 5.4 was utilized to generate risk of bias tables and summaries (Higgins JPT, 2024).

The overall certainty of evidence was evaluated using the GRADE approach (Grading of Recommendations, Assessment, Development, and Evaluations) (Balsheim et al., 2011; Puhan et al., 2014). Although all included studies were RCTs, which initially suggest high certainty, evidence was downgraded based on the following domains: risk of bias, inconsistency, indirectness, imprecision, and publication bias. The certainty of evidence for each outcome—whether from direct, indirect, or network estimates—was classified as high, moderate, low, or very low (Brignardello-Petersen et al., 2018).

For continuous outcomes, standardized mean differences (SMDs) with 95% confidence intervals (CIs) were computed. Heterogeneity between studies was assessed, and potential intransitivity was evaluated by comparing study designs across treatment nodes. Imprecision of the estimates was judged based on whether the 95% CIs crossed pre-specified minimal clinically important differences (Mustafa et al., 2024). Inconsistency was evaluated by examining the magnitude and direction of effects across different comparison paths. Substantial

heterogeneity was defined as an  $I^2$  statistic  $>50\%$  (Guyatt et al., 2023). All evidence evaluations were summarized using GRADEpro GDT (Guideline Development Tool) in adherence to the GRADE framework (Santesso et al., 2020).

## 2.5. Data synthesis and analysis

Both two-arm and multi-arm trials were eligible for inclusion, allowing for the integration of direct comparative evidence within trials and indirect evidence across the network (Jansen et al., 2011). Network meta-analysis (NMA) utilized evidence graphs to visualize all pairwise comparisons, grounded in graph theory. NMA was performed using graph-theoretical methods, where nodes represent interventions and edges represent direct comparisons between them; edge thickness corresponds to the number of available studies (Rücker, 2012).

A Bayesian hierarchical model fitted via Markov chain Monte Carlo (MCMC) methods was employed using either fixed or random effects, depending on the connectedness of the network (Wiksten, Hawkins, Piepho & Gsteiger, 2020). Convergence was assessed using trace plots and density plots. Between-study heterogeneity was estimated using prior distributions (Turner, Jackson, Wei, Thompson & Higgins, 2015). If substantial heterogeneity was detected, sensitivity or subgroup

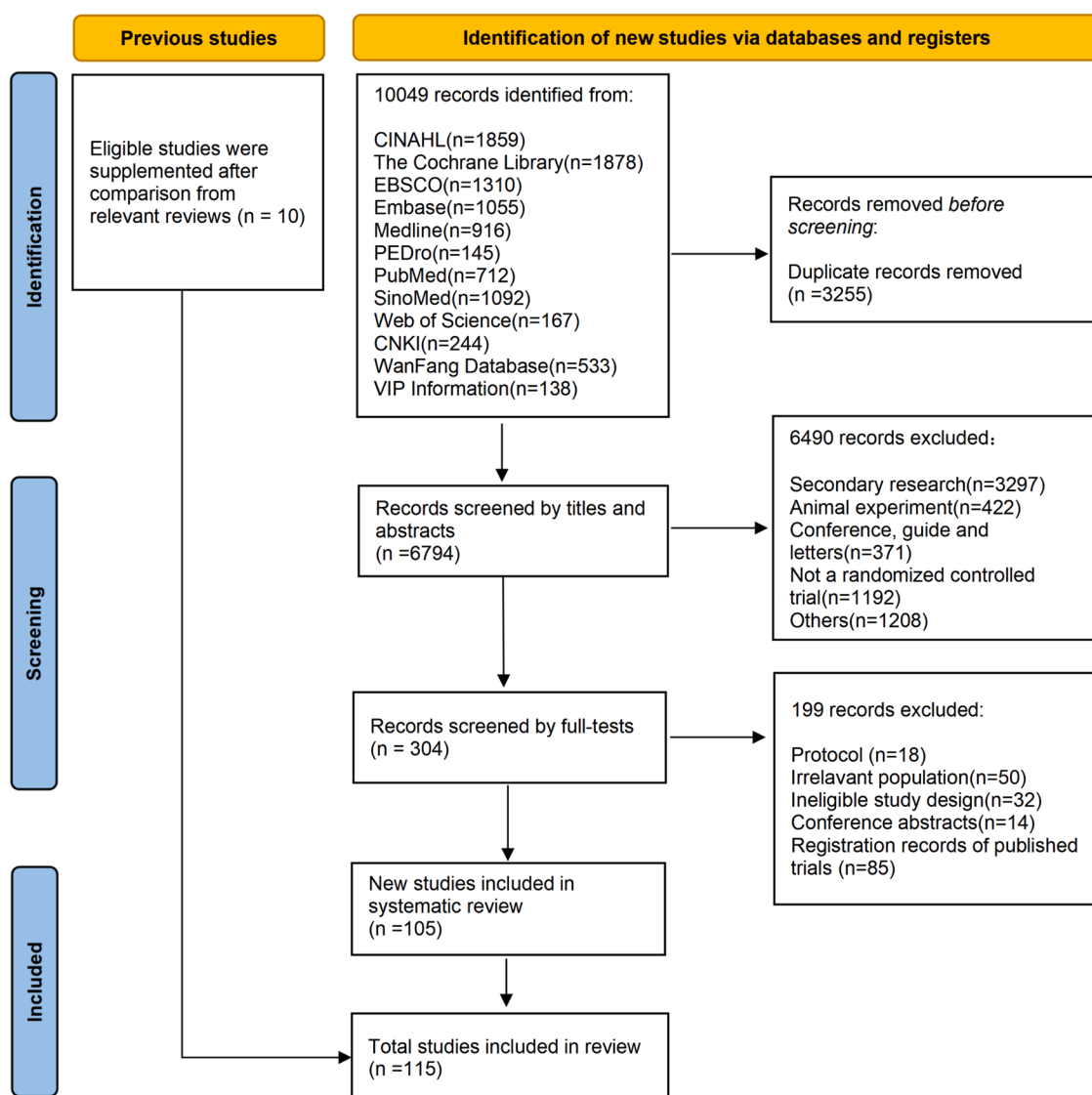


Fig. 1. PRISMA flow diagram of the study selection process.

analyses were planned. Inconsistency between direct and indirect evidence was evaluated using the node-splitting method (van Valkenhoef, Dias, Ades & Welton, 2015).

Treatment rankings were derived using the surface under the cumulative ranking curve (SUCRA), where values range from 0% to 100% (Salanti, Ades & Ioannidis, 2011). Publication bias was assessed using Egger's test and presented via funnel plots (Egger, Smith, Schneider & Minder, 1997). All analyses were conducted in R version 4.5.0 (Neupane, Richer, Bonner, Kibret & Beyene, 2014), with package support from JAGS version 4.3.1 for Bayesian modeling. Results are reported as mean  $\pm$  standard deviation, and treatment effects are expressed as SMDs with 95% CIs.

For the primary outcomes, subgroup analyses followed by meta-regression were planned to explore potential sources of heterogeneity (Higgins JPT, 2024), based on available data, across the following categories:

- (1) Age: <60, 60–69, 70–80, >80 years;
- (2) Intervention duration: <12, 12–16, 17–24, >24 weeks;
- (3) Diagnostic criteria: AWGS, EWGSOP, or other;
- (4) Sex distribution: 0%, 1–49%, 50–99%, 100% male;
- (5) Sample size: categorized by median values;
- (6) Risk of bias: low, some concerns, high.

To statistically test the moderating effects of these categorical variables on the pooled effect sizes, we conducted random-effects meta-regression analyses (Thompson & Higgins, 2002) using the restricted maximum-likelihood (REML) estimator. For each outcome and moderator, we reported Cochran's Q statistic for the model ( $Q_M$ ), its degrees of freedom ( $df$ ), and the associated p-value ( $p$ ). A  $p < 0.05$  was considered indicative of a statistically significant moderating effect. Finally, sensitivity analyses were performed to assess the robustness of the primary outcome results.

### 3. Results

#### 3.1. Search results

Our initial search identified 10,049 citations. After removing 3255 duplicates, 6794 records underwent title and abstract screening, resulting in 304 articles for full-text assessment. Following full-text review, 105 studies met the eligibility criteria. Additionally, 10 studies were included from reference lists of relevant reviews and meta-analyses. Ultimately, a total of 115 RCTs were incorporated into our study.

#### 3.2. Study characteristics

A total of 10,967 people participated in these 115 eligible trials, and their mean age was  $74.85 \pm 8.70$  years (weighted average based on available data from the studies), and the female-to-male ratio was 6021:3807. Nineteen studies exclusively enrolled female participants, while nine studies included only males. The median intervention duration was 12 weeks (mean: 19.4 weeks). Sarcopenia was diagnosed using EWGSOP in 13.8% of studies, AWGS criteria in 39.6%, while others utilized relevant expert consensus or adhered to definitions established by various sources. Study population were almost evenly derived from community and hospital setting, with only three studies conducted in nursing homes.

The specific interventions evaluated in this study were detailed in Supplementary Table S1. Each intervention was systematically categorized and summarized for analysis. To ensure clarity in the graphical presentation of results, abbreviated labels were assigned to each intervention. We provide the complete list of intervention codes alongside their full names for every outcome measure in Supplementary Table S7.

#### 3.3. Risk of bias and grade

The risk of bias assessment revealed clear patterns across domains (Fig. 2). Performance bias was the primary methodological concern, with approximately 75% of studies judged as high risk, largely due to the inherent difficulty of blinding in hands-on nursing interventions. In contrast, selection bias related to random sequence generation was generally well-addressed ( $\approx 75\%$  low risk). Allocation concealment, however, was frequently unclear ( $\approx 65\%$ ). Detection bias showed a mix of low and unclear ratings. Attrition and reporting biases were minimal, with the vast majority ( $\geq 90\%$ ) of studies at low risk. Overall, while the risk of bias was low in reporting and attrition domains, the high prevalence of performance bias highlights a key limitation when interpreting the intervention effects.

The certainty of evidence for each outcome was assessed using the GRADE framework. Due to serious concerns regarding risk of bias and inconsistency ( $I^2 > 50\%$ ), the evidence was downgraded by two levels for key comparisons. Consequently, the overall certainty ranged from moderate to very low across outcomes, with no evidence rated as high certainty. The certainty of evidence for the comparison between resistance training and usual care is presented in Table 1. Supplementary Table S4 presents the GRADE evidence profiles for the key outcome of muscle strength based on direct comparisons against usual care or placebo.

### 4. Primary outcome

#### 4.1. Muscle strength

The network meta-analysis of muscle strength included 87 trials (8780 participants). The network graph comprised 29 nodes, nutrition, nutrition combined with exercise, and resistance training were the most frequently studied interventions.

MCMC chains demonstrated good convergence, with trace plots and density plots indicated stable posterior distributions. In most intervention comparisons, no significant inconsistency was observed between direct and indirect evidence ( $P > 0.05$ ). Overall heterogeneity was relatively low, though some comparisons showed elevated heterogeneity. The following interventions improved muscle strength compared to usual care through the double comparison of league table and probability line chart: endurance followed by resistance training (SMD 2.59, 95% CI 1.50 to 3.69; low certainty), resistance followed by endurance training (SMD 1.49, 95% CI 0.54 to 2.45; low certainty), resistance and cognitive training (SMD 1.35, 95% CI 0.69 to 2.00; low certainty), nutrition combined with aerobic and resistance training (SMD 1.25, 95% CI 0.64 to 1.87; low certainty), aerobic exercise combined with resistance training (SMD 0.43, 95% CI 0.11 to 0.75; moderate certainty) and resistance training alone (SMD 0.42, 95% CI 0.29 to 0.54; very low certainty). Other interventions showed little or no effect relative to usual care or placebo. The network plot, the cumulative ranking curves, and the forest plots of direct pairwise comparisons are presented in Fig. 3, Fig. 4, and Fig. 5, respectively. Related supplementary analyses and extended data are detailed in the Appendix.

#### 4.2. Lean mass

Eighty-two trials (8191 participants) were included. The network graph included 27 nodes; nutrition, nutrition with exercise, and resistance training were the most common. MCMC chains demonstrated good convergence, with both trace plots and density plots indicated stable posterior distributions. Heterogeneity was very high in some direct comparisons ( $I^2 > 90\%$ ), while other comparisons showed low heterogeneity. The inconsistency analysis indicated generally consistent direct and indirect evidence for most comparisons ( $P > 0.05$ ). Through the double comparison of league table and probability line chart, the following interventions increased lean mass compared to usual care:

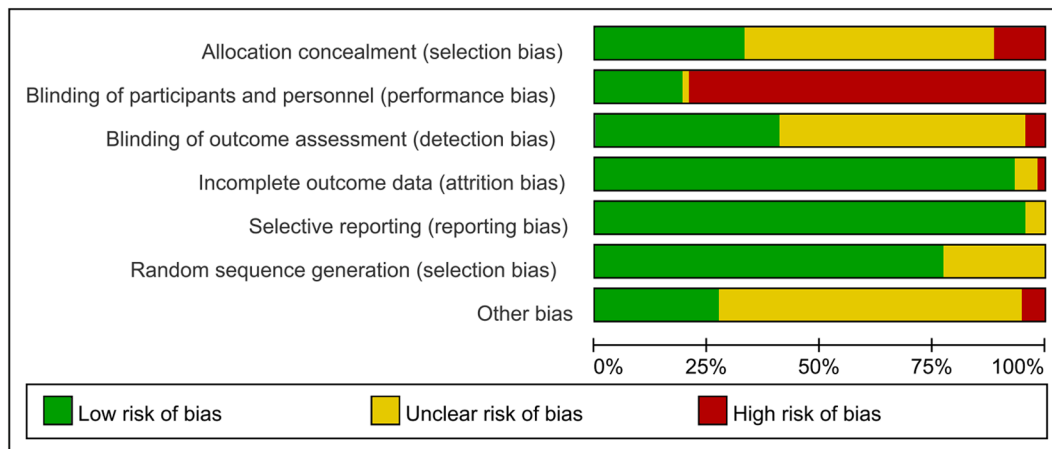


Fig. 2. Risk of bias summary for included studies.

Table 1

GRADE evidence profile: resistance training versus usual care.

| Certainty assessment |                                              |                           |                      |                      |                      |        | N° of patients |     | Effect<br>Absolute<br>(95% CI)                     | Certainty                 |
|----------------------|----------------------------------------------|---------------------------|----------------------|----------------------|----------------------|--------|----------------|-----|----------------------------------------------------|---------------------------|
| N°                   | Study design                                 | Risk of bias              | Inconsistency        | Indirectness         | Imprecision          | Others | RT             | UC  |                                                    |                           |
| 14                   | Muscle strength<br>randomised trials         | very serious <sup>a</sup> | not serious          | serious <sup>b</sup> | not serious          | none   | 403            | 382 | SMD 0.6 SD higher<br>(0.38 higher to 0.83 higher)  | ⊕○○○<br>Very low<br>a,b   |
| 15                   | Lean mass<br>randomised trials               | very serious <sup>a</sup> | serious <sup>c</sup> | not serious          | not serious          | none   | 556            | 542 | SMD 0.51 SD higher<br>(0.24 higher to 0.79 higher) | ⊕○○○<br>Very low<br>a,c   |
| 15                   | Physical performance<br>randomised trials    | serious <sup>a</sup>      | serious <sup>d</sup> | not serious          | serious <sup>e</sup> | none   | 428            | 448 | SMD 0.28 SD higher<br>(0.11 lower to 0.67 higher)  | ⊕○○○<br>Very low<br>a,d,e |
| 7                    | Blood biomarker<br>randomised trials         | serious <sup>a</sup>      | not serious          | not serious          | not serious          | none   | 128            | 146 | SMD 0.27 SD higher<br>(0.02 higher to 0.51 higher) | ⊕⊕⊕○<br>Moderate<br>a     |
| 13                   | Fat mass<br>randomised trials                | serious <sup>a</sup>      | not serious          | not serious          | not serious          | none   | 253            | 280 | SMD 0.56 SD lower<br>(0.74 lower to 0.38 lower)    | ⊕⊕⊕○<br>Moderate<br>a     |
| 6                    | Quality of life<br>randomised trials         | serious <sup>a</sup>      | not serious          | not serious          | serious <sup>e</sup> | none   | 119            | 116 | SMD 0.12 SD higher<br>(0.28 lower to 0.51 higher)  | ⊕⊕○○<br>Low<br>a,e        |
| 2                    | Knee extension strength<br>randomised trials | serious <sup>a</sup>      | serious <sup>f</sup> | not serious          | not serious          | none   | 47             | 31  | SMD 0.69 SD higher<br>(0.23 higher to 1.14 higher) | ⊕⊕○○<br>Low<br>a,f        |

a. Blinding and allocation concealment were assessed as having a high risk of bias in some studies.

b. Some trials included populations with concomitant diseases, which may limit generalizability.

c. High heterogeneity was observed ( $I^2 = 77.6\%$ ,  $\tau^2 = 0.29$ ).d. Substantial heterogeneity was present ( $I^2 = 86.8\%$ ,  $\tau^2 = 0.52$ ).

e. The 95% CI is wide and includes the null value.

f. Considerable heterogeneity was detected ( $I^2 = 75.8\%$ ,  $\tau^2 = 0.32$ ).

general remote training (SMD 1.35, 95% CI 0.41 to 2.29; low certainty), endurance followed by resistance training (SMD 1.35, 95% CI 0.41 to 2.29; low certainty), and nutrition with aerobic and resistance training (SMD 1.25, 95% CI 0.64 to 1.87; moderate certainty). Additionally, resistance and cognitive training (SMD 1.35, 95% CI 0.69 to 2.00; low certainty), resistance followed by endurance training (SMD 1.05, 95% CI 0.13 to 1.97; low certainty), aerobic exercise (SMD 1.77, 95% CI 1.21 to 2.33; very low certainty), and rehabilitation exercise (SMD 1.77, 95% CI 1.26 to 2.28; very low certainty) increased lean mass compared to usual care. In comparison with placebo, TCM (SMD 1.18, 95% CI 0.70 to 1.65; very low certainty), nutrition and exercise (SMD 0.39, 95% CI 0.28 to

0.50; low certainty), and nutrition supplementation (SMD 0.23, 95% CI 0.05 to 0.41; moderate certainty) increased lean mass. Other interventions showed minimal effects.

#### 4.3. Physical performance

Eighty-one trials (8569 participants) were included. The network graph included 27 nodes. MCMC chains demonstrated good overall convergence. Heterogeneous varied across comparisons, necessitating sensitivity analysis or subgroup analysis for some. Inconsistency was generally low ( $P > 0.05$ ). The following interventions improved physical

**Table 2**  
Abbreviations of Interventions.

|       |                                                         |
|-------|---------------------------------------------------------|
| WBVT  | whole-body vibration training                           |
| AE    | aerobic exercise                                        |
| RT    | resistance training                                     |
| MT    | multicomponent training                                 |
| WM    | western medicine                                        |
| NTBI  | nursing theory-based intervention                       |
| RE    | rehabilitation exercise                                 |
| UC    | usual care                                              |
| TCM   | traditional Chinese medicine                            |
| NE    | nutrition and exercise                                  |
| RCT   | resistance and cognitive training                       |
| N     | nutrition                                               |
| BS-RT | biomyoelectric stimulation and resistance training      |
| NART  | nutrition combined with aerobic and resistance training |
| HE    | health education                                        |
| HE-P  | health education and protein                            |
| GRT   | general remote training                                 |
| ART   | AI-based remote training                                |
| FTT   | face-to-face traditional training                       |
| M-E   | medicine and exercise                                   |
| P-E   | placebo and exercise                                    |
| E + R | endurance followed by resistance training               |
| R + E | resistance followed by endurance training               |

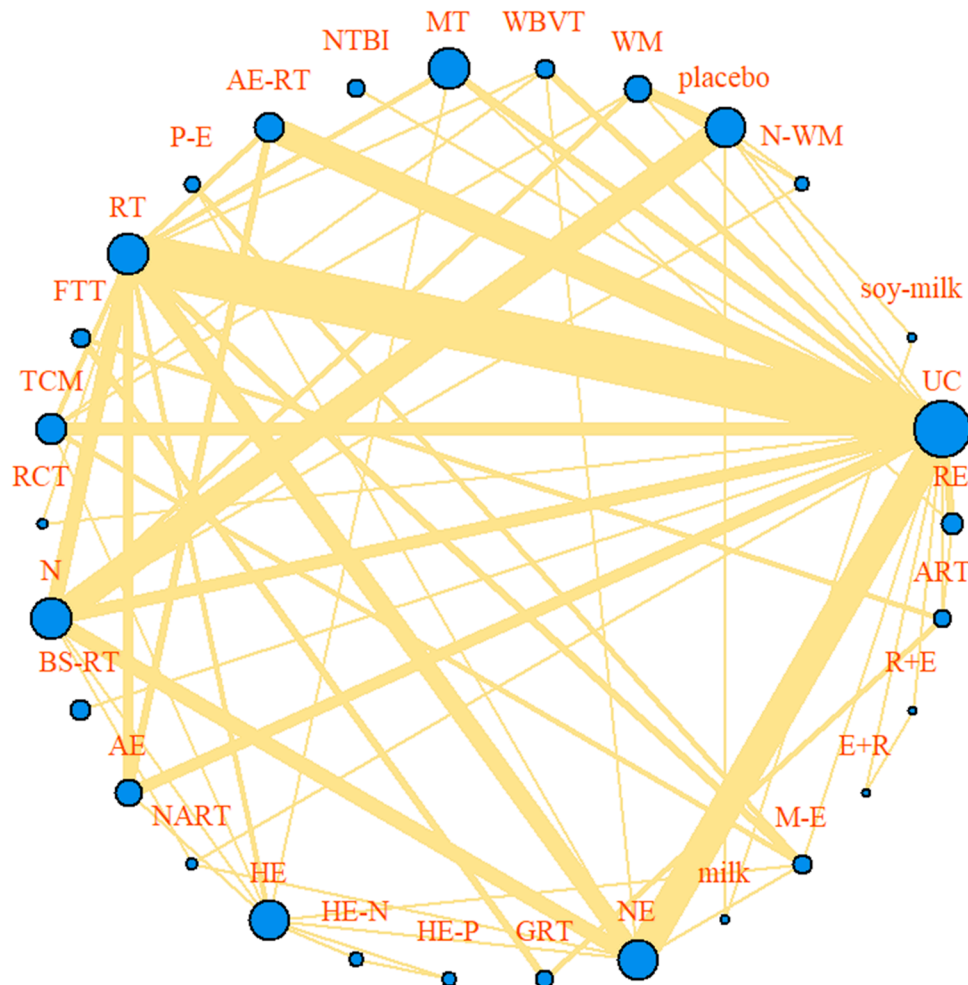
performance compared to usual care: general remote training (SMD 2.39, 95% CI 1.26 to 3.53; low certainty), nutrition combined with aerobic and resistance training (SMD 1.73, 95% CI 1.09 to 2.37; very low certainty), and AI (Artificial Intelligence) -based remote training (SMD

1.93, 95% CI 1.19 to 2.68; low certainty). Resistance and cognitive training (SMD 0.99, 95% CI 0.35 to 1.62; very low certainty) and aerobic exercise (SMD 0.97, 95% CI 0.45 to 1.50; low certainty) also improved physical performance compared to usual care. Milk (SMD 1.59, 95% CI 0.67 to 2.51; very low certainty), TCM (SMD 1.02, 95% CI 0.61 to 1.43; very low certainty), and nutrition supplementation (SMD 0.88, 95% CI 0.48 to 1.28; very low certainty) improved physical performance compared to placebo. Other interventions showed negligible effects.

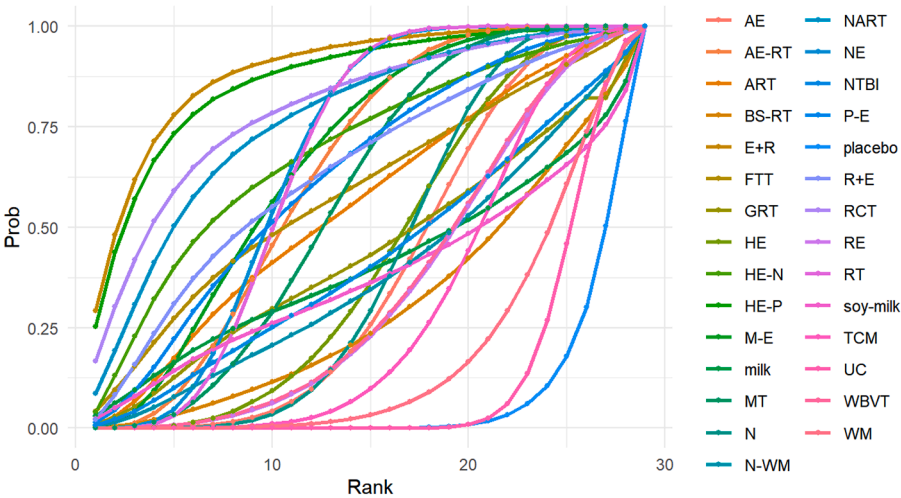
**5. Secondary outcome**

**5.1. Fat mass**

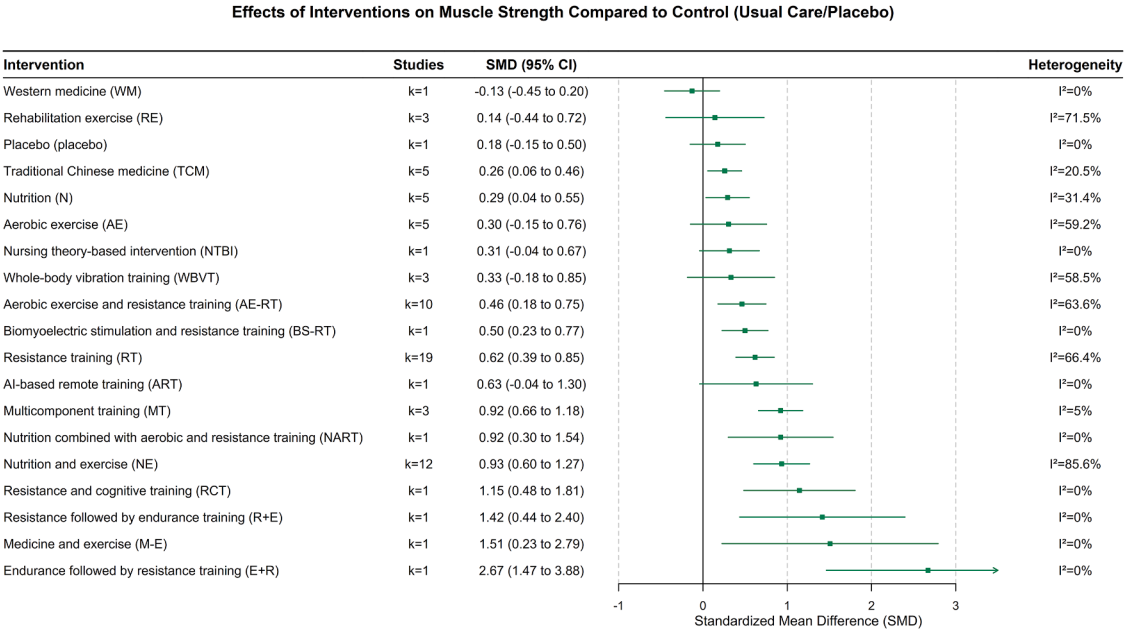
Thirty-seven trials (2031 participants) were included. The network graph included 19 nodes; resistance training was the most studied intervention. Heterogeneity was high in partial direct comparisons. Inconsistency was generally low ( $P > 0.05$ ). The following interventions reduced body fat mass compared to usual care: resistance training (SMD  $-0.57$ , 95% CI  $-0.75$  to  $-0.40$ ; moderate certainty), aerobic exercise and resistance training (SMD  $-0.32$ , 95% CI  $-0.56$  to  $-0.09$ ; moderate certainty), and aerobic exercise (SMD  $-0.48$ , 95% CI  $-0.87$  to  $-0.09$ ; low certainty). TCM (SMD  $-0.53$ , 95% CI  $-0.97$  to  $-0.08$ ; very low certainty) reduced body fat mass compared to placebo. Other interventions showed little effects.



**Fig. 3.** Network plot of interventions for muscle strength.  
Notes: Network plot showing direct comparisons between interventions. Node size is proportional to the number of participants. Lines represent direct comparisons, with thickness proportional to the number of trials. See Table 2 for abbreviations.



**Fig. 4.** Cumulative ranking probability plot for muscle strength.  
Notes: Cumulative ranking probabilities showing the probability of each intervention being ranked at each position (1 = best). See Table 2 for abbreviations.



**Fig. 5.** Effects of interventions on muscle strength: direct comparisons with control.  
Notes: This figure presents only direct comparisons against a common control; head-to-head comparisons between active interventions are not shown. Each row represents a pooled estimate from a random-effects meta-analysis. The size of the data marker corresponds to the weight of the study in the meta-analysis. The diamond represents the overall pooled estimate. SMD > 0 favors the intervention. Heterogeneity is presented as I<sup>2</sup>.

5.2. Quality of life

Thirty-seven trials (2840 participants) were included. The network graph included 19 nodes. Heterogeneity was high in some direct comparisons. Inconsistency was generally low ( $P > 0.05$ ). The following interventions improved quality of life compared to usual care: whole-body vibration training (SMD 1.04, 95% CI 0.20 to 1.88; low certainty). Nutrition combined with exercise (SMD 0.78, 95% CI 0.53 to 1.03; low certainty) and aerobic exercise (SMD 0.71, 95% CI 0.53 to 1.03; low certainty) probably improved quality of life compared with usual care. Other interventions showed subtle effects.

5.3. Blood biomarker

Twenty-nine trials (2155 participants) were included. These studies assessed a range of circulating biomarkers, primarily encompassing pro-inflammatory cytokines (e.g., IL-6, IL-1 $\beta$ , IL-18, TNF- $\alpha$ ), acute-phase reactants (CRP), and growth factors (e.g., IGF-1, GDF-8), among others. The network graph included 16 nodes; resistance training and nutrition were the most studied frequently interventions. Heterogeneity was high in partial direct comparisons. Inconsistency was generally low ( $P > 0.05$ ). The following interventions reduced blood inflammatory markers compared to usual care: aerobic exercise (SMD -1.98, 95% CI -2.31 to -1.66; low certainty), resistance training (SMD -1.69, 95% CI

–1.95 to –1.43; low certainty) and aerobic exercise with resistance training (SMD –1.81, 95% CI –2.01 to –1.62; moderate certainty). Other interventions showed marginal effects.

#### 5.4. Knee extension strength

Twenty-one trials (1342 participants) were included. The network graph included 13 nodes; resistance training was the most studied interventions. Heterogeneity was variable across comparisons. Inconsistency was generally low ( $P > 0.05$ ). The following interventions increased knee extension strength compared to usual care: rehabilitation exercise (SMD 0.56, 95% CI 0.09 to 1.03; low certainty) and resistance training (SMD 0.55, 95% CI 0.33 to 0.76; low certainty). TCM (SMD 0.73, 95% CI 0.00 to 1.46; very low certainty) increased knee extension strength compared to placebo. Other interventions showed minimal effects.

### 6. Subgroup, meta-regression, and sensitivity analyses

#### 6.1. Subgroup analyses

Subgroup analyses revealed differential intervention effects across key moderators (Fig. 6). Adults aged  $> 80$  years showed the largest improvement in physical performance (SMD=0.76, 95% CI: 0.50–0.96,  $P < 0.001$ ) and muscle strength (SMD=0.77,  $P = 0.017$ ). Regarding intervention duration, shorter-term interventions ( $< 12$  weeks) yielded the strongest effects on physical performance (SMD=0.85) and lean mass (SMD=0.93), with lean mass also responding sustainably in longer-term studies ( $>24$  weeks, SMD=0.64). Small-sample studies ( $\leq 10$  participants) consistently reported inflated effect sizes across outcomes, suggesting small-study bias. Studies with a high risk of bias generally yielded larger effect estimates, and their exclusion meaningfully reduced pooled effect sizes. Diagnostic criteria significantly moderated

muscle strength outcomes, with EWGSOP criteria associated with lower effects (SMD=–0.85,  $P = 0.015$ ). No significant effects were observed across gender subgroups.

#### 6.2. Meta-Regression analyses

To assess potential ecological bias and the statistical significance of moderators identified in subgroup analyses, meta-regression with Cochran's Q test was performed (Supplementary Table S5). The results indicated that sample size grouping significantly moderated muscle strength ( $Q_M=8.10$ ,  $df=2$ ,  $p = 0.017$ ) and lean mass ( $Q_M=7.21$ ,  $df=2$ ,  $p = 0.027$ ), corroborating subgroup findings. Age grouping ( $Q_M=10.43$ ,  $df=4$ ,  $p = 0.034$ ) and gender ratio ( $Q_M=8.33$ ,  $df=3$ ,  $p = 0.040$ ) significantly moderated lean body mass only, suggesting outcome-specific demographic influences. Intervention duration significantly moderated physical performance alone ( $Q_M=11.36$ ,  $df=4$ ,  $p = 0.023$ ), aligning with subgroup trends. Neither diagnostic criteria nor risk of bias showed significant moderating effects across any outcome ( $p > 0.05$ ).

#### 6.3. Sensitivity analyses

Sensitivity analysis revealed that after excluding studies with a high risk of bias, the effect sizes for muscle strength (SMD decreased from 0.50 to 0.12,  $p > 0.05$ ) and physical performance (SMD decreased from 0.31 to –0.09,  $p > 0.05$ ) were substantially reduced to non-significance, while the effect size for lean body mass also diminished (SMD decreased from 0.34 to 0.06,  $p > 0.05$ ). These findings warrant cautious interpretation: the small number of low-risk studies ( $k = 3-5$ ) may have limited statistical power, and functional outcomes are particularly susceptible to subjective expectations, which may explain why high-risk studies tended to overestimate effects. Results were robust to the choice of statistical model or heterogeneity estimator. However, using only one arm per multi-arm study rendered the physical performance

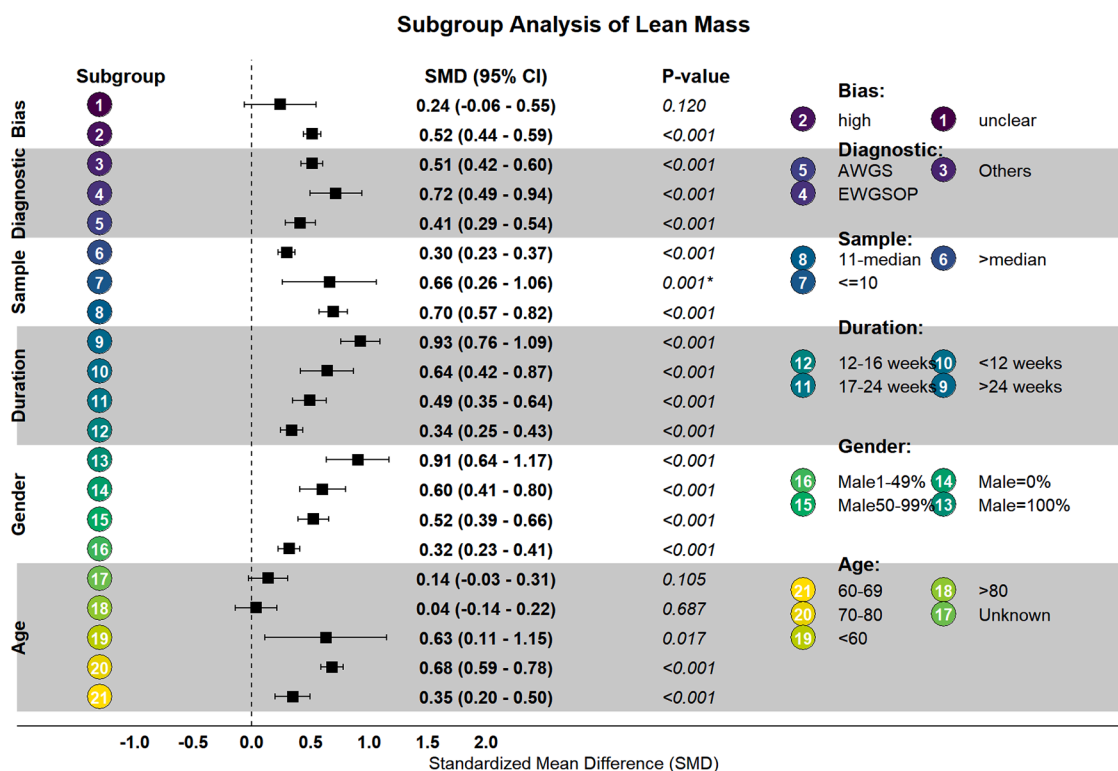


Fig. 6. Forest plot of subgroup analysis for lean mass.

Notes: This forest plot presents the SMDs with 95% CIs from subgroup analyses of interventions aimed at improving lean mass in patients with sarcopenia. Each row represents a different subgroup category and its levels. The vertical line at SMD = 0 indicates no effect.

outcome non-significant (SMD 0.29, 95% CI: -0.05 to 0.62). Leave-one-out analysis revealed no study had a disproportionate influence. The results of these sensitivity analyses are summarized in Supplementary Table S6.

## 7. Discussion

### 7.1. Primary findings

Regarding the primary outcomes of sarcopenia, our studies found that intervention can be broadly categorized by training content and intervention methods. Although studies on remote training were relatively limited, it ranked highly for both lean mass and physical performance, suggesting its potential as an effective intervention for managing sarcopenia outside traditional healthcare settings. However, this finding should be interpreted with caution. The evidence base for remote training was limited, with only two studies included, increasing the risk of insufficient statistical power and chance findings. Notably, although the effect sizes were large, the wide CIs indicate substantial imprecision in the estimates. The observed large effects may be attributable to the high standardization of remote interventions and the capacity for real-time monitoring of adherence, which could enhance intervention fidelity compared to unsupervised home exercise. This aligns with emerging evidence from recent randomized controlled trials of digital interventions for other conditions. For instance, AI-assisted remote rehabilitation (Yao et al., 2026) and smartphone-based platforms (Netz et al., 2025) enabling remote assessment and home-based personalized exercise have been shown to effectively deliver individualized programs for older adults. These findings underscore a clear opportunity for future research to integrate and rigorously evaluate digital approaches within sarcopenia intervention frameworks. While promising, these signals require confirmation through high-quality randomized controlled trials before definitive recommendations can be made.

In terms of intervention components, resistance exercise combined with other exercise modalities or nutritional supplementation showed significant benefits on primary outcomes. For secondary outcomes, both resistance training and aerobic exercise demonstrated beneficial effects, though resistance training proved more effective in interventions targeting body fat mass and inflammatory blood markers. From a clinical perspective, reductions in body fat mass are relevant because excess adiposity, particularly intramuscular fat, can exacerbate insulin resistance and impair muscle quality, contributing to the development of sarcopenic obesity (Li et al., 2022). Similarly, improvements in inflammatory markers such as CRP are clinically meaningful, as chronic low-grade inflammation is a key driver of muscle catabolism and functional decline in aging populations (Tuttle, Thang & Maier, 2020). These findings support the notion that optimal sarcopenia management may require addressing multiple interrelated pathologies simultaneously, including adiposity and systemic inflammation.

However, nutritional supplementation alone demonstrated limited effects on sarcopenia. This is also consistent with the findings of other researchers. The composition and ratios of nutrients, and how they are integrated with specific exercise modalities, will be a central research focus. Meanwhile, TCM may improve lean mass and physical performance, though the supporting evidence was of very low certainty. Investigating how interventions influence these measures remains an important aspect of sarcopenia research.

Our subgroup and meta-regression analyses clarified the substantial heterogeneity and contextualized the findings for primary outcomes. Clinically, effects were strongest in adults aged  $\geq 80$  years and with interventions shorter than 12 weeks, highlighting a high-priority population and a potentially efficient treatment timeframe. Methodologically, small-study effects and a high risk of bias consistently inflated effect sizes, suggesting the pooled estimates may overstate the absolute benefit. Low methodological quality in research is a major source of bias. Since the effectiveness of interventions relies heavily on

evidence quality, future research should prioritize minimizing bias to ensure more reliable and valid conclusions. Consequently, while the comparative efficacy rankings appear robust, the overall magnitude of benefit should be interpreted with caution due to methodological limitations in the available evidence.

### 7.2. Comparisons with other studies

Previous studies have consistently affirmed the central role of resistance training in the management of sarcopenia (Fu, Wang, Zhao, Li & Song, 2025; Sánchez et al., 2025). Expanding on this established foundation, our network meta-analysis further reveals that the most effective strategy is not resistance training in isolation, but rather its integration with either aerobic exercise or nutritional support (Zhang, Yang, Wang & Zhang, 2025). In the present study, nutritional interventions involving whole foods were categorized as “nutrition”, while encapsulated supplements were classified as “medicine”. Medicine did not show significant effects in our analysis, which may be related to age-related differences in absorption and metabolism among older adults. It is noteworthy, however, that earlier research (Gu, Zhang, Wu & Wang, 2025; Wang & Lei, 2025) has reported that certain standalone supplements, such as  $\beta$ -hydroxy- $\beta$ -methylbutyrate (HMB) and probiotics, improved muscle strength. This aligns with other findings indicating that protein supplementation alone, in the absence of exercise, yields limited benefits (Yoshimura, Matsumoto, Inoue, Okamura & Kuzuya, 2025). Importantly, any nutritional strategy must be tailored to an individual’s physiological status and comorbidities, as various chronic conditions can impose dietary restrictions (Abuelazm et al., 2025).

Our findings also provide novel insights into the underlying pathophysiology and emerging management approaches for sarcopenia. The ranking outcomes from our network meta-analysis suggest that aerobic or resistance exercise constitutes the most effective non-pharmacological strategy for ameliorating inflammatory markers, thereby directly addressing a core mechanism in sarcopenia pathogenesis (Feng et al., 2025). Furthermore, this study offers preliminary network estimates for TCM interventions. Although research in this area remains limited by a lack of standardized methodologies (He et al., 2025), the observed potential of TCM underscores the need for further rigorous investigation. Future studies should prioritize large-scale, methodologically robust clinical trials and explore precision management strategies guided by biomarkers, such as inflammatory profiles.

### 7.3. Strengths and limitations

As an exhaustive systematic review and network meta-analysis, this study provides a comprehensive benchmarking of all available interventions for sarcopenia. Its principal strength lies in the rigorous methodological framework employed to ensure robust and valid comparisons. We conducted extensive sensitivity analyses, subgroup analyses, and meta-regression to validate the network geometry and to quantify the influence of key methodological variables. A broad spectrum of sarcopenia-relevant outcomes was evaluated; mortality was not analyzed due to its infrequent reporting and lack of significant association in existing trials. Our findings extend beyond clinical implications to underscore fundamental methodological challenges in the field, where factors such as sample size, blinding, and study design critically influence the evidence base. More fundamentally, the persistently limited long-term effectiveness of interventions compels a re-evaluation of management strategies, favoring multimodal approaches that target the complex, underlying drivers of progression in geriatric syndromes.

Despite these strengths, this study also identifies limitations inherent to its methodology and the current landscape of sarcopenia intervention research. First, the robustness of our findings is compromised by the suboptimal quality of included studies. Methodological limitations were most pronounced in performance bias (only 20% low risk for blinding of

participants) and detection bias (only 40% low risk for blinding of outcome assessment). These biases tend to inflate effect estimates, particularly for subjective outcomes. Our sensitivity analysis confirmed this: excluding high-risk studies rendered muscle strength and physical performance non-significant, while attenuating lean mass. This suggests that low-quality studies substantially inflated the observed benefits. Functional outcomes, being more susceptible to bias due to their subjective components, showed greater attenuation, whereas lean mass was relatively less affected. These findings align with our subgroup and meta-regression analyses indicating that study quality moderates effect estimates. Thus, the true intervention effects may be smaller than overall estimates suggest, warranting cautious interpretation. Second, although we explored heterogeneity, inherent differences in study populations and intervention details may compromise the transitivity assumption underlying the network meta-analysis, introducing potential bias into indirect comparisons. Third, for certain interventions, the evidence network remains sparse and relies heavily on indirect comparisons, limiting the certainty of these efficacy rankings. Additionally, current sarcopenia intervention research lacks breakthrough innovations, with most effective strategies still focusing on conventional approaches such as nutrition exercise combinations.

## 8. Conclusion

Research on sarcopenia is ongoing and evolving. Our study indicates that resistance exercise remains the primary exercise intervention for sarcopenia, while combined intervention is more significant for promotion, such as the resistance training combined with aerobic exercise or nutrition. In the future, remote intervention will become dominant. Although single nutrition and single medicine were not proven to have a very high effect in this study, nutritional intake plays a significant role in the health of the elderly. If TCM can be accepted as a treatment method, it will become an effective intervention approach.

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## CRediT authorship contribution statement

**Jianting Cao:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yan Liu:** Investigation, Formal analysis, Data curation. **JiETING Hu:** Validation, Supervision. **Tiantian Zhai:** Methodology. **Ziping Pang:** Validation, Investigation, Data curation. **Rao Li:** Methodology. **Fangping Dang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.archger.2026.106239](https://doi.org/10.1016/j.archger.2026.106239).

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